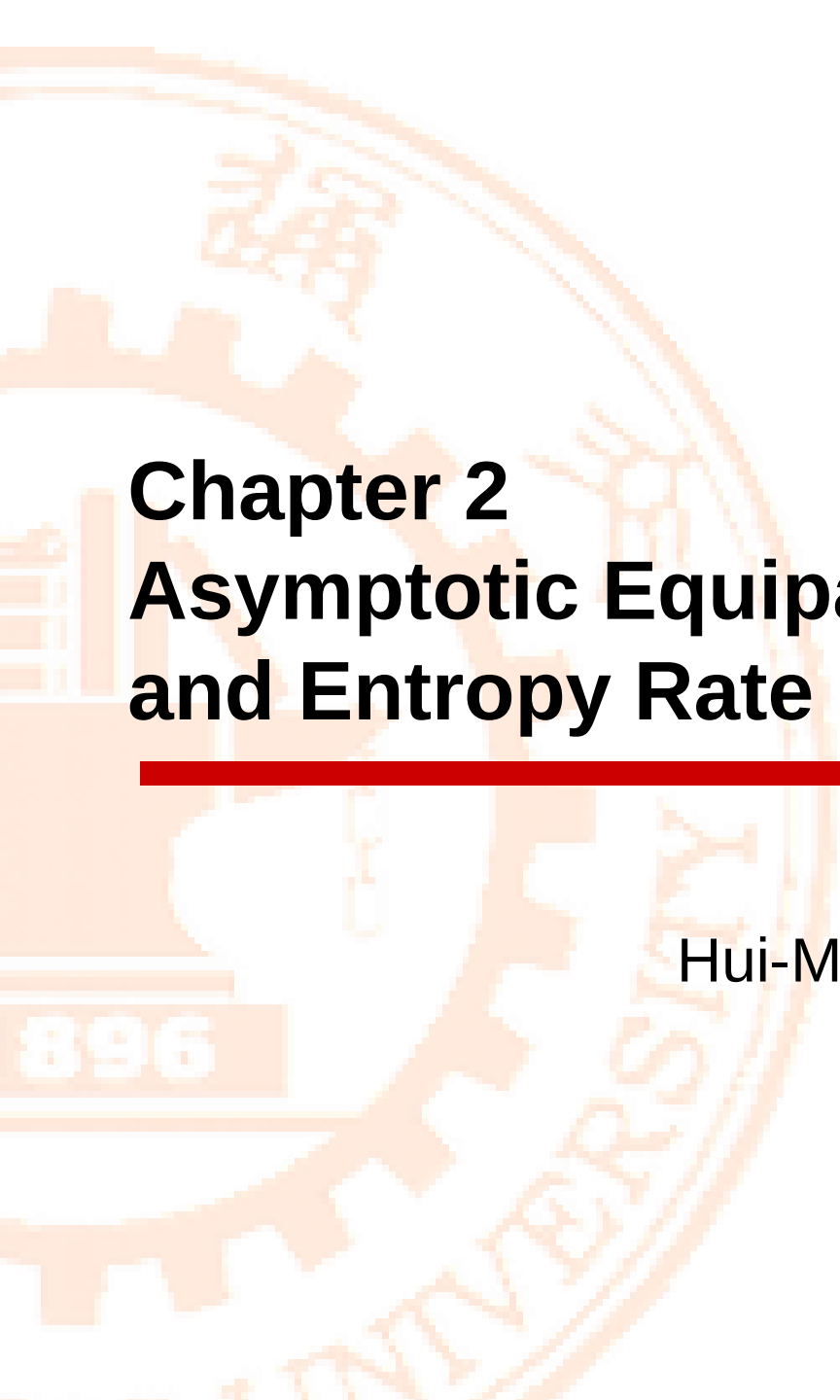




Chapter 2

Asymptotic Equipartition Property and Entropy Rate (渐进等分性质和熵率)

Hui-Ming Wang



Preview

- In Chapter 1, we introduced many basic notions
 - Entropy
 - Joint Entropy and Conditional Entropy
 - Mutual Information
- These notions are used to describe some features of a random variable, or relationship of several random variables.
随机变量或多个随机变量之间的关系
- But how to apply these basic notions to communications? (Note that we have mentioned that we are considering the information theory in communications) 如何在通信中应用?
- In this chapter, we use a discrete random variable X with probability density function (pdf) $p(X)$ to describe a **source**. The outputs of the source is the realizations of the discrete random variable X . 用随机变量来描述信源



Preview

□ Source examples

■ Character

ASCII: A~Z, 0~9, ... 128

GB2312-800: 6763,

GBK: GB2312+BIG5, more than 30000

■ Sound

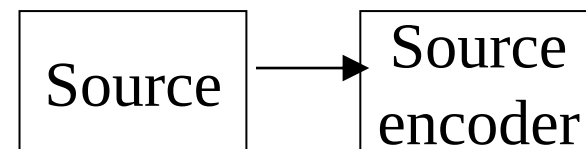
Voice: PCM,

Music: MP3.

■ Image/Video

Picture: BMP, JPG,

Movie: MPEG1(VCD), MPEG2(DVD), RMVB



message

- Let us consider how to describe the source by information bits more efficiently. We first consider the independent, identically distributed (i.i.d.) random variables case. (独立同分布)



Question

- Suppose that we have a source generates independent binary information bits sequence randomly, 0100111010110...
- If $0 \rightarrow 0.55, 1 \rightarrow 0.45$, then as the length of the sequence goes very large, what kind of sequence we are most likely to see ?
010011011001110001111011001100110101110010101 ?
1111111110101111100111110101110101011111111111 ?
- If $0 \rightarrow 0.2, 1 \rightarrow 0.8$, ?
- If $0 \rightarrow 0.1, 1 \rightarrow 0.9$
010011011001110001111011001100110101110010101 ?
1111111110101111100111110101110101011111111111 ?

How many bits we **actually** need to describe the sequences generated by such a source (or the random variable X) ??



Outline

- Convergence of Random Variables
- Laws of Large Numbers
- Asymptotic Equipartition Property (AEP)
- Properties
- Entropy Rates of a Stochastic Process
- AEP for Stationary Stochastic Processes



Asymptotic Equipartition Property



Laws of Large Numbers

□ **Weak** Law of Large Numbers (弱大数定理)

Let $X_1, X_2, \dots, X_n$ be a sequence of independent and identically distributed random variables, each having finite mean $E[X_i] = \mu$
Then, for any $\varepsilon > 0$

$$P\left(\left|\frac{X_1 + X_2 + \dots + X_n}{n} - \mu\right| \geq \varepsilon\right) \xrightarrow{n \rightarrow \infty} 0$$

□ **Strong** Law of Large Numbers (强大数定理)

Let $X_1, X_2, \dots, X_n$ be a sequence of independent and identically distributed random variables, each having finite mean $E[X_i] = \mu$
Then,

$$P\left(\lim_{n \rightarrow \infty} \frac{X_1 + X_2 + \dots + X_n}{n} - \mu = 0\right) = 1$$



Converge *in probability*
(依概率收敛)



Weak Law of Large Numbers
(弱大数定理)

Converge *almost surely/
with probability 1*
(几乎处处收敛)



Strong Law of Large Numbers
(强大数定理)



Laws of Large Numbers

- Differences between the weak law and the strong law
 - The *weak law* states that for a specified large n , the average $\bar{X}_n$ is likely to be near μ . Thus, it leaves open the possibility that $|\bar{X}_n - \mu| > \varepsilon$ happens an **infinite number of times**, although at infrequent intervals.
 - 概率趋向于0，但是实际发生的次数仍然可能是无穷大
 - The *strong law* shows that this almost surely will not occur. In particular, it implies that with **probability 1**, we have that for any $\varepsilon > 0$ the inequality $|\bar{X}_n - \mu| < \varepsilon$ holds for all large enough n .



Asymptotic Equipartition Property (AEP)

□ Intuitive idea of AEP

- Assume X a R.V. so that $P(X=1) = p = 1 - P(X=0)$ We observe a sequence of R. V. $X_1, X_2 \cdots X_n$ with $P(X_i) = P(X)$

$$P(X_1, X_2 \cdots X_n = x_1, x_2 \cdots x_n) = p^{\sum x_i} (1-p)^{n - \sum x_i}$$

- We can see these sequences are not equally likely 不等概, some sequences have high probability to appear. However, as $n \rightarrow \infty$, only $\approx 2^{nH(X)}$ sequences appear, and all of them appear with the same probability $\approx 1/2^{nH(X)}$
- The typical set 典型集 is the set of all these $\approx 2^{nH(X)}$ sequences. Each called a typical sequence 典型序列.
- Since these are the only sequences that appear, we do not care about the other ones. $\rightarrow$ we just need $nH(X)$ bits to represent all original sequences instead of n bits,



AEP

□ AEP

■ AEP:

Let $X_1, X_2 \dots X_n$ be a sequence of i.i.d random variables with distribution $p(X)$

$$\Rightarrow -\frac{1}{n} \log p(X_1, X_2 \dots, X_n) \xrightarrow{n \rightarrow \infty} H(X), \text{ in probability}$$

■ Proof: using weak Law of Large Numbers

Let $X_1, X_2, \dots X_n$ be a sequence of i. i. d random variables with distribution $p(X)$

$$\frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{n \rightarrow \infty} E[X], \text{ in probability}$$

Corollary:
$$\frac{1}{n} \sum_{i=1}^n f(X_i) \xrightarrow{n \rightarrow \infty} E[f(X)], \text{ in probability}$$



AEP

□ Proof:

$$-\frac{1}{n} \log p(X_1, X_2, \dots, X_n) = -\frac{1}{n} \sum_i \log p(X_i)$$

$\rightarrow -E \log p(X)$ in probability
 $= H(X),$

Corollary

$$\frac{1}{n} \sum_{i=1}^n f(X_i) \xrightarrow{n \rightarrow \infty} E[f(X)], \text{ in probability}$$

“AEP即为大数定理”



AEP

■ Typical set

Definition The *typical set* $A_\epsilon^{(n)}$ with respect to $p(x)$ is the set of sequences $(x_1, x_2, \dots, x_n) \in \mathcal{X}^n$ with the property

$$2^{-n(H(X)+\epsilon)} \leq p(x_1, x_2, \dots, x_n) \leq 2^{-n(H(X)-\epsilon)}.$$

或者 $A_\epsilon^{(n)} = \left\{ (x_1, x_2, \dots, x_n) : 2^{-n(H(X)+\epsilon)} < p(x_1, x_2, \dots, x_n) < 2^{-n(H(X)-\epsilon)} \right\}$

或者 $A_\epsilon^{(n)} = \left\{ (x_1, x_2, \dots, x_n) : \left| -\frac{1}{n} \log p(x_1, x_2, \dots, x_n) - H(X) \right| < \epsilon \right\}$

■ Interpretation

□ Observe sequence $(x_1, x_2, \dots, x_n)$, if $n \rightarrow \infty$, we will see $(x_1, x_2, \dots, x_n) \in A_\epsilon^{(n)}$, $\forall \epsilon > 0$ and $p(x_1, x_2, \dots, x_n) \simeq 2^{-nH(X)}$, as $\epsilon \rightarrow 0$

We will see $|A_\epsilon^{(n)}| \simeq 2^{nH(X)}$

随着观察序列的增长, 我们只能看到典型序列出现, 并且各自的概率接近等概。



AEP

□ Properties of typical set 典型集合的性质1-4

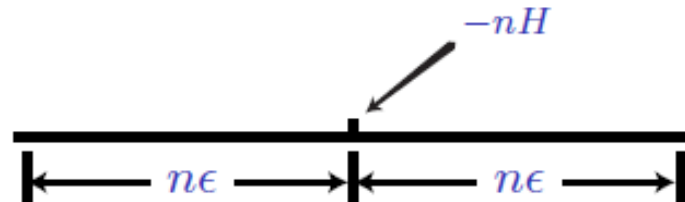
1. If $(x_1, x_2, \dots, x_n) \in A_{\epsilon}^{(n)}$, then

$$H(X) - \epsilon \leq -\frac{1}{n} \log p(x_1, x_2, \dots, x_n) \leq H(X) + \epsilon.$$

Proof: It is immediate from the definition of typical set $A_{\epsilon}^{(n)}$

$$A_{\epsilon}^{(n)} = \left\{ (x_1, x_2, \dots, x_n) : 2^{-n(H(X)+\epsilon)} < p(x_1, x_2, \dots, x_n) < 2^{-n(H(X)-\epsilon)} \right\}$$

Typical set are those sequences with log probability within the range





AEP

2. $\Pr\{A_\epsilon^{(n)}\} > 1 - \epsilon$ for n sufficiently large.

Proof: from the definition of AEP $-\frac{1}{n}\log p(x_1, x_2, \dots, x_n) \xrightarrow{p} H(X)$

Definition: $\forall \epsilon > 0, \delta > 0, \exists n_0 = N, \forall n > n_0$

$$P\left(\left|-\frac{1}{n}\log p(X_1, X_2, \dots, X_n) - H(X)\right| < \epsilon\right) > 1 - \delta$$
$$\Rightarrow P(A_\epsilon^{(n)}) > 1 - \epsilon, \text{ if take } \delta = \epsilon$$

The typical set has probability nearly 1

只有典型序列才会出现

$$\Pr\{(X_1, X_2, \dots, X_n) : p(X_1, X_2, \dots, X_n) = 2^{-n(H \pm \epsilon)}\} \approx 1$$



AEP

3. $|A_\epsilon^{(n)}| \leq 2^{n(H(X)+\epsilon)}$, where $|A|$ denotes the number of elements in the set A .

$$\begin{aligned} 1 = \sum_{\mathbf{x} \in \mathcal{X}^n} p(\mathbf{x}) &\geq \sum_{\mathbf{x} \in A_\epsilon^{(n)}} p(\mathbf{x}) \geq \sum_{\mathbf{x} \in A_\epsilon^{(n)}} 2^{-n(H(X)+\epsilon)} \\ &= 2^{-n(H(X)+\epsilon)} |A_\epsilon^{(n)}|, \end{aligned}$$

The number of elements in the typical set is a bit less than $2^{nH(X)}$

典型序列总数稍小于 $2^{nH(X)}$



AEP

4. $|A_\epsilon^{(n)}| \geq (1 - \epsilon)2^{n(H(X) - \epsilon)}$ for n sufficiently large.

$$1 - \epsilon < \Pr\{A_\epsilon^{(n)}\} \leq \sum_{\mathbf{x} \in A_\epsilon^{(n)}} 2^{-n(H(X) - \epsilon)} = 2^{-n(H(X) - \epsilon)} |A_\epsilon^{(n)}|$$

$$\leq \sum_{\mathbf{x} \in A_\epsilon^{(n)}} p(\mathbf{x})$$

典型序列总数稍大于 $2^{nH(X)}$



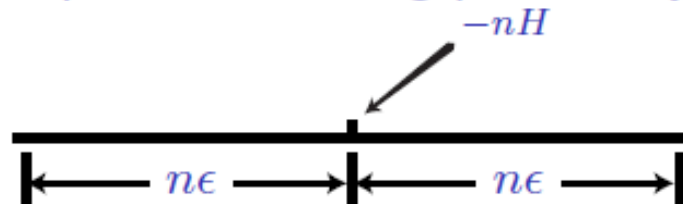
AEP

- From property 3 and 4, if n sufficiently large, $\varepsilon \rightarrow 0$

$$|A_{\varepsilon}^{(n)}| \simeq 2^{nH(X)}$$

- From property 2, if n sufficiently large and I observe sequence $(x_1, x_2 \cdots x_n) \in A_{\varepsilon}^{(n)}$ with probability 1.
- From property 1, All sequences in typical set have the same probability $\simeq 2^{-nH(X)}$

Typical set are those sequences with log probability within the range





AEP

- Assume X a R.V. so that $P(X=1) = p < \frac{1}{2}$

$$H(X) = -p \log p - (1-p) \log(1-p)$$

- The typical set $A_\epsilon^{(n)}$ is the set of sequence with **about** pn ones and $(1-p)n$ zeros.
- The probability of a typical sequence is about $p^{pn}(1-p)^{(1-p)n} = 2^{-nH(X)}$
- The number of sequence with pn ones is $C_n^{pn} = \frac{n!}{(pn)!(n-pn)!}$
- There are totally 2^n sequences, but most of them are collectively very improbable.

Example $P(X=1) = 0.1, P(X=0) = 0.9 \quad H(X) \approx 0.469$

$$n = 100 \quad 2^{100} \approx 10^{30} \quad 2^{nH} = 10^{14} \ll 10^{30}$$

So the number of typical sequences is much smaller than the

number of possible sequences. 典型序列的数量远小于序列总数



AEP

- Note that the most probable sequence have almost all zeros, but they are not in the typical set. ---- there are not enough of them to matter. (概率最大的那个序列是全零序列, 但是其不在典型集中)

Most likely sequence has probability $0.9^{100} \approx 2.66 \times 10^{-5}$
but a typical sequence has probability 7.62×10^{-15}

There are **exponentially many sequences** that are typical than there are "high" probability sequences.

The **probability of each individual sequence** goes to zero, but the **size of the set of typical sequences** grows fast enough $n \rightarrow \infty$



AEP

□ High Probability Sets （高概率集合）

- Typical set is a fairly small set that contains most of the probability.
- It is not clear whether it is the smallest such set.
- Smallest set

Definition For each $n = 1, 2, \dots$, let $B_\delta^{(n)} \subset \mathcal{X}^n$ be the smallest set with

$$\Pr\{B_\delta^{(n)}\} \geq 1 - \delta.$$

- The smallest set is different from the typical set, but must have significant intersection with and therefore must have about as many elements. （两者定义不同，但是含有的元素个数基本一样，且存在显著交集）

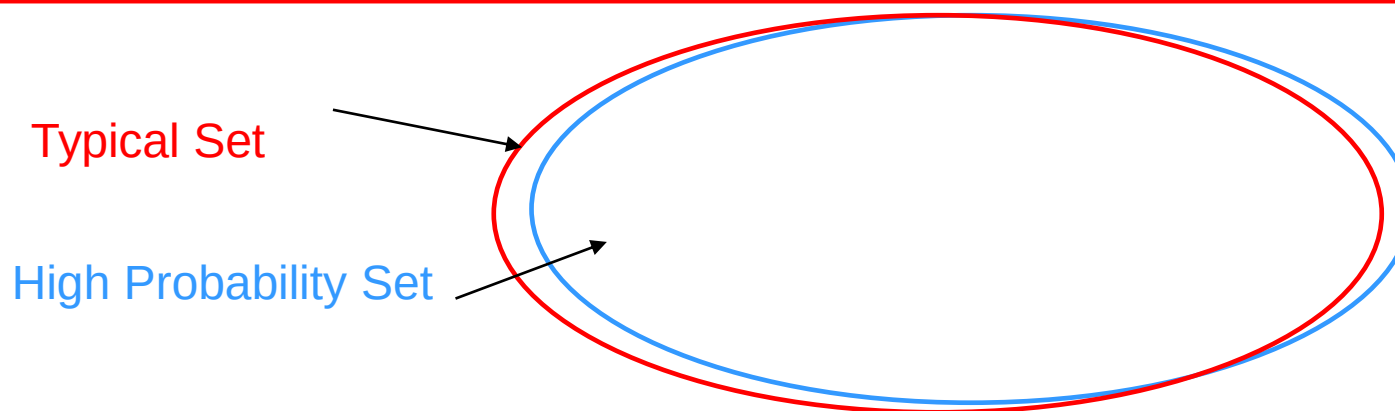


AEP

Theorem 3.3.1 Let $X_1, X_2, \dots, X_n$ be i.i.d. $\sim p(x)$. For $\delta < \frac{1}{2}$ and any $\delta' > 0$, if $\Pr\{B_\delta^{(n)}\} > 1 - \delta$, then

$$\frac{1}{n} \log |B_\delta^{(n)}| > H - \delta' \quad \text{for } n \text{ sufficiently large.}$$

$B_\delta^{(n)}$ must have at least 2^{nH} elements, is about the same size as the $A_\epsilon^{(n)}$

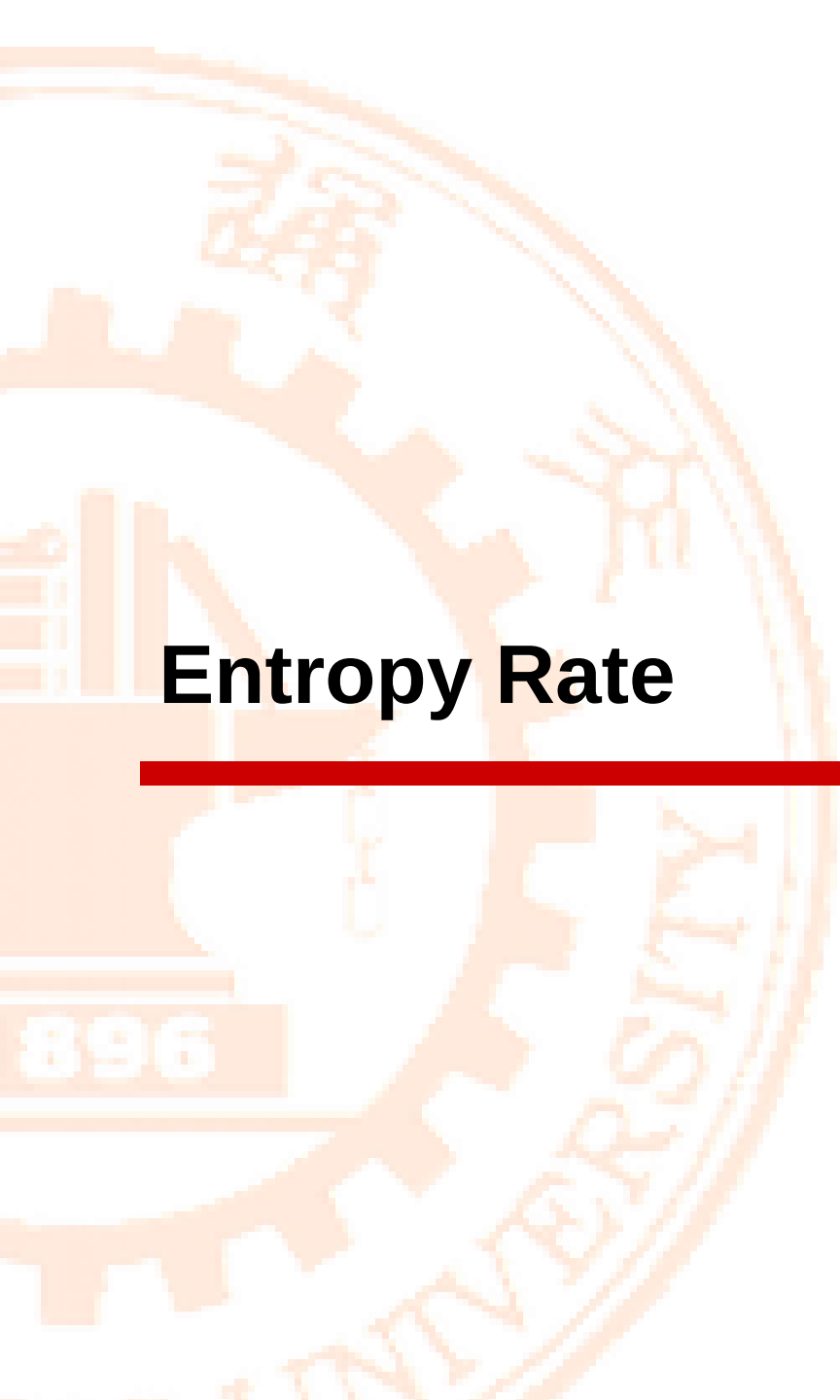




AEP

- The AEP establishes that $nH(X)$ bits suffice on the average to describe n **independent identically distributed** random variables. (前面讨论的是独立同分布随机变量的情况)
- But what if the random variables are dependent? In particular, what if the random variables form **a stationary process**?

In the following, we will show, just as in the i. i. d. case, that the entropy $H(X_1, X_2, \dots, X_n)$ grows (asymptotically) linearly with n at a rate $H(\mathcal{X})$, which we will call the **entropy rate of the stochastic (random) process**. (随机过程的熵率)



Entropy Rate



Stochastic (Random) Process

□ Stochastic process (随机过程)

- Definition: a stochastic process is just a time sequence of random variables.
- $\{X_i\}_{i \in I}$ or $X(t)$, at each n or t , is a random variable.
- To describe a stochastic process, we use the **joint distribution**

$$P(X_1 = x_1, X_2 = x_2 \cdots X_n = x_n)$$

□ Stationary (平稳)

- A stochastic process is stationary when the joint probability mass of any subset of random variables is invariant with respect to shifts in the time index. (严平稳定义)

$$P(X_1 = x_1, X_2 = x_2 \cdots X_n = x_n) = P(X_{1+i} = x_1, X_{2+i} = x_2 \cdots X_{n+i} = x_n)$$

- The stationary property makes a stochastic process easy to analyze → time index is not needed.



Stochastic Process

□ Ergodic Property (各态历经)

- A stationary process is ergodic if it satisfies the law of large numbers

Mean:

均值

$$\frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{n \rightarrow \infty} E[X], \text{ with probability } 1$$

Time average

时间平均

Ensemble average (calculated in prob. space)

集平均

**Auto-correlation
Function**

自相关函数

$$\frac{1}{n} \sum_{i=1}^n X_i X_{i+k} \xrightarrow{n \rightarrow \infty} E[X_i X_{i+k}], \text{ with probability } 1$$

Time average

Ensemble average (calculated in prob. space)



Entropy Rates of a Stochastic Process

□ Entropy Rate

- Measure of how $H(X_1, X_2, \dots, X_n)$ **grow** with n . (Rate of growth 增长率)

Definition The *entropy* of a stochastic process $\{X_i\}$ is defined by

$$H(\mathcal{X}) = \lim_{n \rightarrow \infty} \frac{1}{n} H(X_1, X_2, \dots, X_n)$$

when the limit exists.

- Simple examples

- i. i. d R. V. $X_1, X_2, \dots, X_n$

$$H(\mathcal{X}) = \lim_{n \rightarrow \infty} \frac{H(X_1, X_2, \dots, X_n)}{n} = \lim_{n \rightarrow \infty} \frac{nH(X_1)}{n} = H(X_1)$$

- Independent but not identically distributed $X_1, X_2, \dots, X_n$

$$H(X_1, X_2, \dots, X_n) = \sum_{i=1}^n H(X_i), \quad \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n H(X_i)$$



Entropy Rates of a Stochastic Process

- identically distributed but not independent $X_1, X_2, \dots, X_n$

$$\begin{aligned} H(\mathcal{X}) &= \lim_{n \rightarrow \infty} \frac{H(X_1, X_2, \dots, X_n)}{n} = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n H(X_i | X_{i-1}, \dots, X_1) \\ &\leq \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n H(X_i) = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n H(X) = H(X) \end{aligned}$$

- Another definition

$$H'(\mathcal{X}) = \lim_{n \rightarrow \infty} H(X_n | X_{n-1}, X_{n-2}, \dots, X_1) \text{ when the limit exists.}$$

Interpretation: Growing of uncertainty when add a new variable.

The first is the per symbol entropy of the n random variables,
The second is the conditional entropy of the last random variable given the past.



Entropy Rates of a Stochastic Process

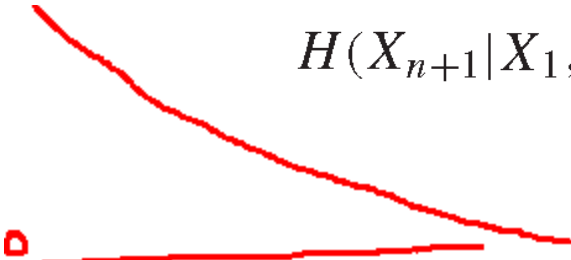
For a stationary stochastic process, the two limits exist and are equal.

$$H(\mathcal{X}) = H'(\mathcal{X}).$$

□ Proof: two steps

■ Limit exists

Conditioning reduces entropy


$$H(X_{n+1}|X_1, X_2, \dots, X_n) \leq H(X_{n+1}|X_n, \dots, X_2)$$

$$= H(X_n|X_{n-1}, \dots, X_1),$$

Stationarity of the process

A decreasing sequence of nonnegative numbers, the limit always exists 一个递减的非负数列，极限总是存在

■ Equality

Entropy Rates of a Stochastic Process

Theorem 4.2.3 (*Cesáro mean*) If $a_n \rightarrow a$ and $b_n = \frac{1}{n} \sum_{i=1}^n a_i$, then $b_n \rightarrow a$.

Formal Proof: Let $\epsilon > 0$. Since $a_n \rightarrow a$, there exists a number $N(\epsilon)$ such that $|a_n - a| \leq \epsilon$ for all $n \geq N(\epsilon)$. Hence,

$$\begin{aligned}
 |b_n - a| &= \left| \frac{1}{n} \sum_{i=1}^n (a_i - a) \right| \\
 &\leq \frac{1}{n} \sum_{i=1}^n |a_i - a| \\
 &\leq \frac{1}{n} \sum_{i=1}^{N(\epsilon)} |a_i - a| + \frac{n - N(\epsilon)}{n} \epsilon \\
 &\leq \frac{1}{n} \sum_{i=1}^{N(\epsilon)} |a_i - a| + \epsilon
 \end{aligned}$$

$$H'(\mathcal{X}) = \lim_{n \rightarrow \infty} H(X_n | X_{n-1}, X_{n-2}, \dots, X_1)$$

$$H(\mathcal{X}) = \lim \frac{H(X_1, X_2, \dots, X_n)}{n}$$

$$= \lim \frac{1}{n} \sum_{i=1}^n H(X_i | X_{i-1}, \dots, X_1)$$



Entropy Rates of a Stochastic Process

- Basic idea: Since most of the terms in the sequence $\{a_k\}$ are eventually close to a , then the average of the first n terms is also eventually close to a

By the chain rule,
$$\frac{H(X_1, X_2, \dots, X_n)}{n} = \frac{1}{n} \sum_{i=1}^n H(X_i | X_{i-1}, \dots, X_1),$$

The entropy rate is the time average of the conditional entropies. The conditional entropies tend to a limit H' .

$$\begin{aligned} H(\mathcal{X}) &= \lim_{n \rightarrow \infty} \frac{H(X_1, X_2, \dots, X_n)}{n} = \lim_{n \rightarrow \infty} H(X_n | X_{n-1}, \dots, X_1) \\ &= H'(\mathcal{X}). \end{aligned} \quad \square$$



Entropy Rates of a Stochastic Process

1	space	0.1817	15	M	0.02105
2	E	0.1073	16	U	0.02010
3	T	0.0856	17	G	0.01633
4	A	0.0668	18	Y	0.01623
5	O	0.0654	19	P	0.01623
6	N	0.0581	20	W	0.01260
7	R	0.0559	21	B	0.01179
8	I	0.0510	22	V	0.00752
9	S	0.0499	23	K	0.00344
10	H	0.04305	24	X	0.00136
11	D	0.03100	25	J	0.00108
12	L	0.02775	26	Q	0.00099
13	F	0.02395	27	Z	0.00063
14	C	0.02260			



Entropy Rates of a Stochastic Process

□ Example

Equally likely

$$H_{\max} = \log 27 \approx 4.755 \text{ bits/symbol}$$

independent

$$H_1(X) = -0.1817\log_2 0.1817 - 0.1073\log_2 0.1073 - \dots - \\ 0.00063\log_2 0.00063 \approx 4 \text{ bits/symbol}$$

Not independent

Combinations: th, he, in, er, an, re, ed, on, es, st, en, at, to, nt, ha, nd, ou, ea, ng, as.

$$H_2(X) \approx 3.3 \text{ bits/symbol}$$



AEP for stationary stochastic processes

- The significance of the entropy rate of a stochastic process arises from the AEP for a stationary ergodic process.
- $\{X_i\}$ a stationary ergodic process

$$-\frac{1}{n} \log P(X_1, X_2 \cdots X_n) \rightarrow H(\{X_i\}), \text{ with probability } 1$$

All the things (properties) we saw for AEP of i.i.d. processes hold, substituting $H(X)$ with $H(\{X_i\})$

when $\varepsilon \rightarrow 0, n \rightarrow \infty$

$$1) \quad |A_\varepsilon^{(n)}| \simeq 2^{nH(\{X_i\})} \qquad 2) \quad P(A_\varepsilon^{(n)}) \rightarrow 1$$

$$3) \quad P(x_1, x_2 \cdots x_n) \simeq 2^{-nH(\{X_i\})}, \text{ if } (x_1, x_2 \cdots x_n) \in A_\varepsilon^{(n)}$$

We can therefore represent the typical sequences of length n using approximately $nH(\{X_i\})$ bits, which shows the significance of the entropy rate as the average description length for a stationary ergodic process.



Summary

□ Asymptotic Equipartition Property (AEP)

AEP. “Almost all events are almost equally surprising.” Specifically, if $X_1, X_2, \dots$ are i.i.d. $\sim p(x)$, then

$$-\frac{1}{n} \log p(X_1, X_2, \dots, X_n) \rightarrow H(X) \text{ in probability.} \quad (3.28)$$

□ Properties

1. If $(x_1, x_2, \dots, x_n) \in A_\epsilon^{(n)}$, then $p(x_1, x_2, \dots, x_n) = 2^{-n(H \pm \epsilon)}$.
2. $\Pr \{A_\epsilon^{(n)}\} > 1 - \epsilon$ for n sufficiently large.
3. $|A_\epsilon^{(n)}| \leq 2^{n(H(X) + \epsilon)}$, where $|A|$ denotes the number of elements in set A .



Summary

□ Entropy Rates of a Stochastic Process

Entropy rate. Two definitions of entropy rate for a stochastic process are

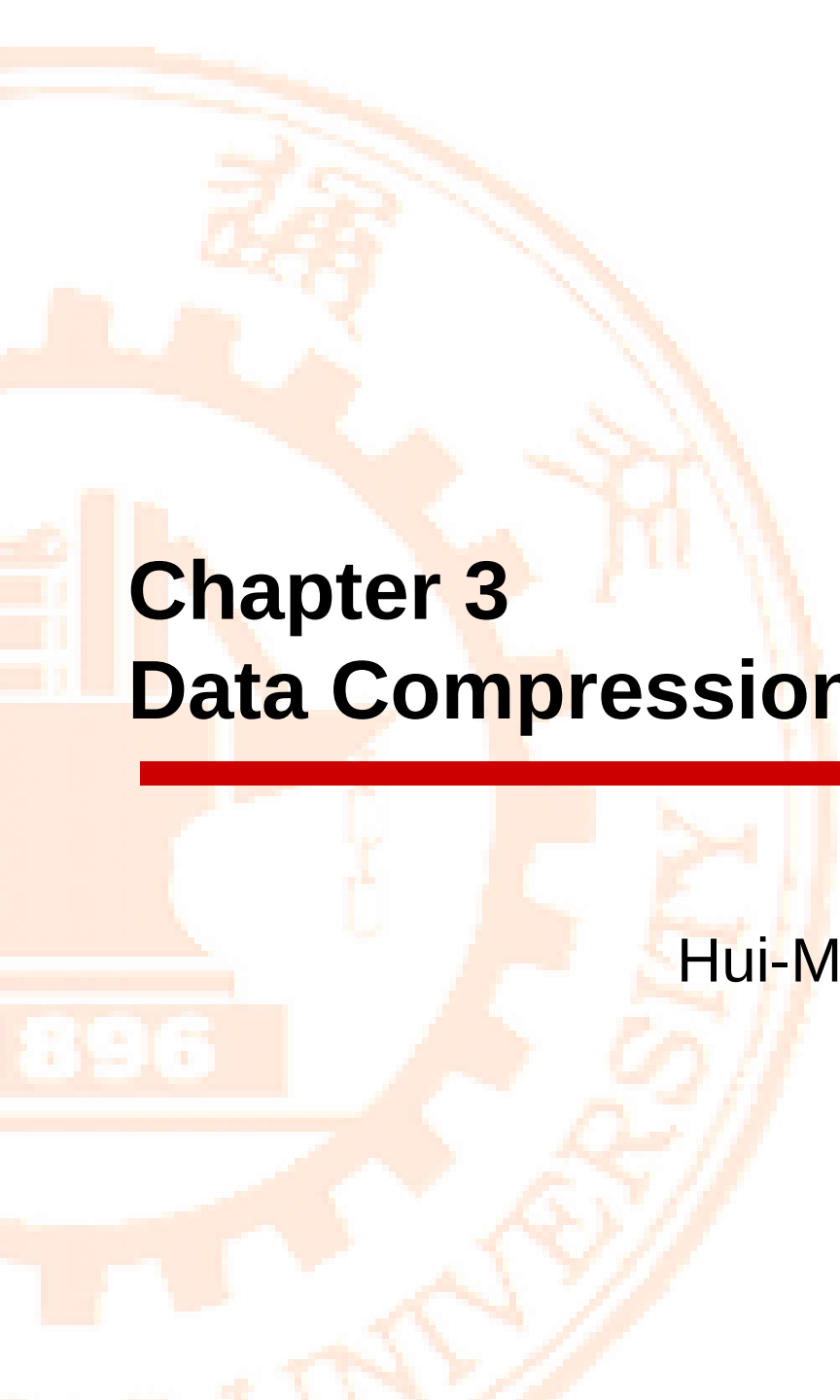
$$H(\mathcal{X}) = \lim_{n \rightarrow \infty} \frac{1}{n} H(X_1, X_2, \dots, X_n), \quad (4.76)$$

$$H'(\mathcal{X}) = \lim_{n \rightarrow \infty} H(X_n | X_{n-1}, X_{n-2}, \dots, X_1). \quad (4.77)$$

For a stationary stochastic process,

$$H(\mathcal{X}) = H'(\mathcal{X}). \quad (4.78)$$

□ AEP for Stationary Stochastic Processes



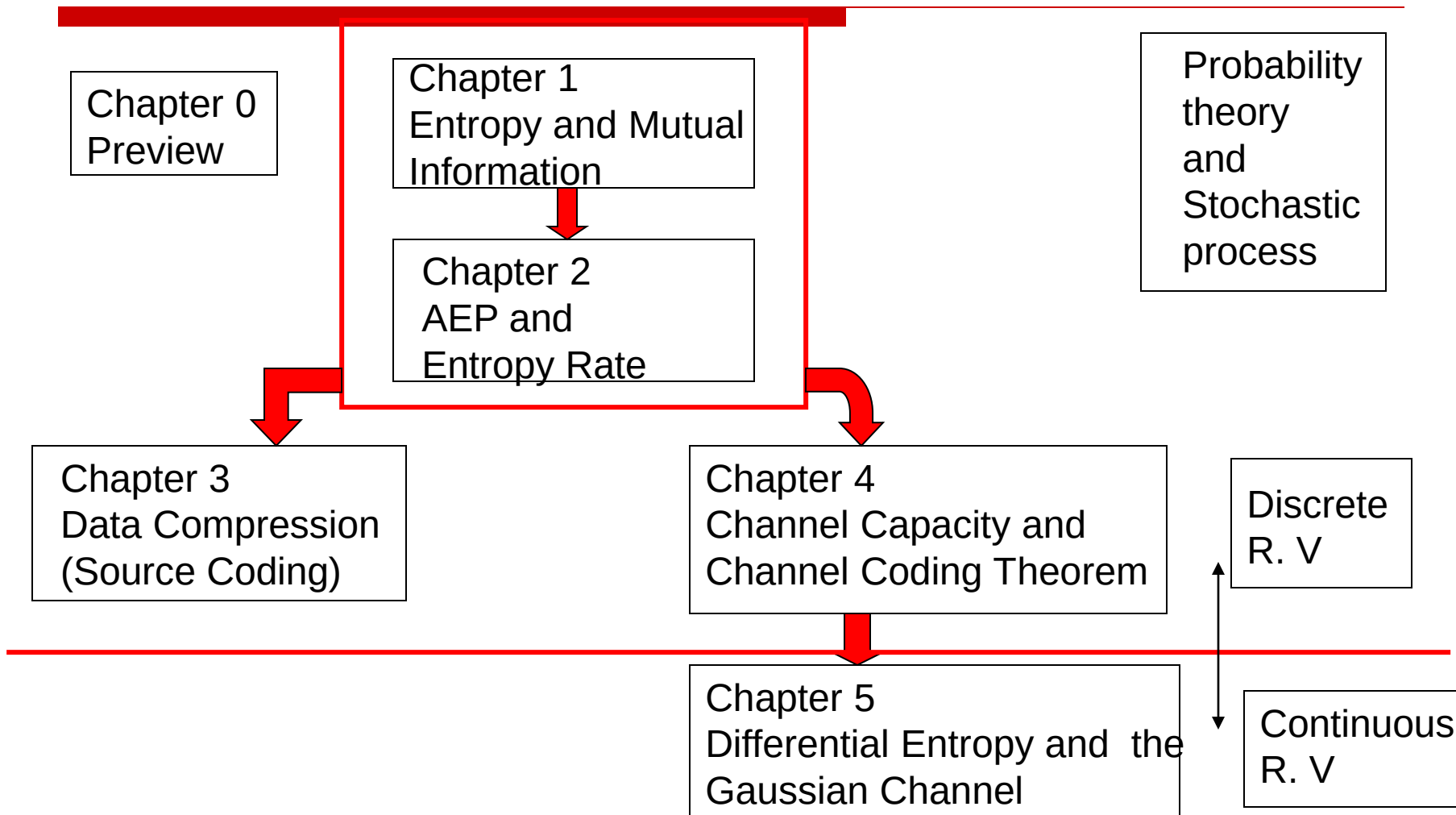
Chapter 3

Data Compression (Source Coding)

Hui-Ming Wang



Structure

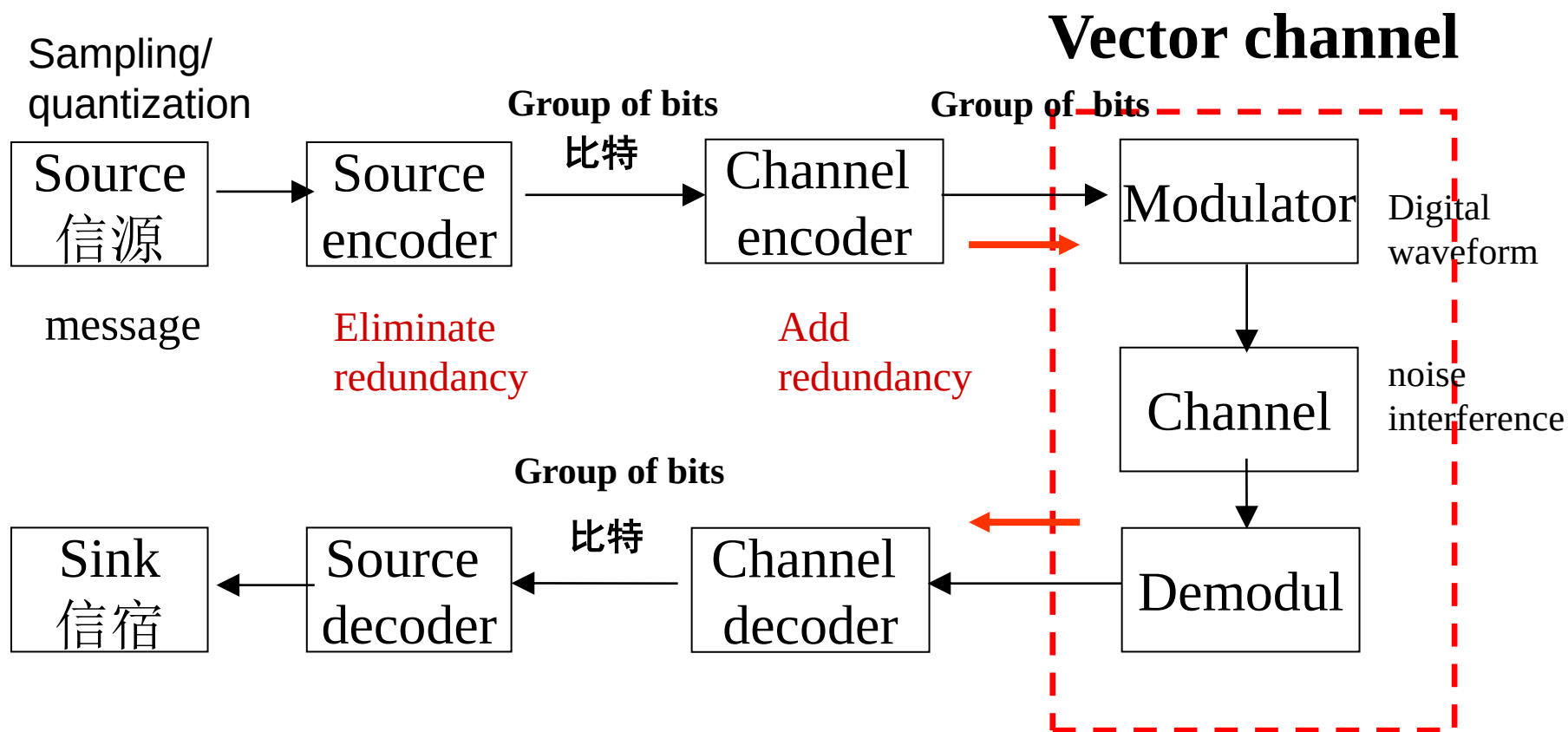




Outline

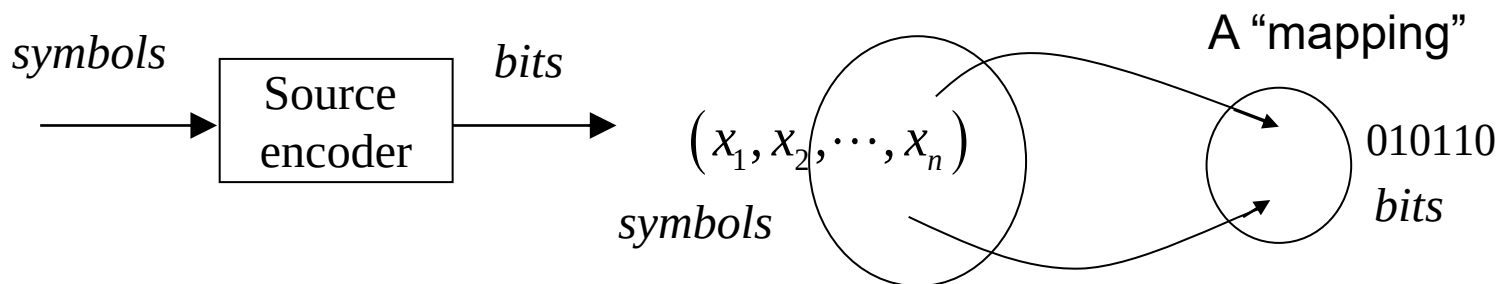
- Introduction
- AEP for Data Compression
- Source Coding
- Kraft Inequality
- Optimal Codes
- An example -- Huffman Code

□ Digital communications system



Introduction

- The source encoder converts the sequence of **symbols** from the source to a sequence of **binary** digits, preferably using **as few binary digits per symbol as possible**.



- Transmit as few bits as possible.—efficiency-- Compression
 - Files $\rightarrow$.rar, .zip (lossless)
 - PCM $\rightarrow$ MP3 (Lossy)
 - BMP $\rightarrow$ JPEG (Lossy)
- The source decoder performs the inverse operation.



Introduction

□ Application of AEP

- For a random source X , the output sequences are not equally likely, some sequences have high probability to appear.
- we donnot care about the other ones. we just need $nH(X)$ bits to represent all original sequences instead of n bits.
- For a binary case, $n \rightarrow nH(X)$ actually is a compression.



AEP for Data Compression



Application of AEP to Data Compression

- Let $X_1, X_2 \dots X_n$ be a sequence of i.i.d random variables with distribution $p(X)$
 $\forall \varepsilon > 0$, we can choose n sufficiently large so that there exists a code that maps sequence $x^n = (x_1, x_2, \dots, x_n)$ of length n into binary strings so that
 - i) The mapping is **one-to-one** (i.e., invertible)
 - ii) $E \left[\frac{1}{n} l(X^n) \right] \leq H(X) + \varepsilon$ with $l(X^n)$ being the length in bits of the binary codeword corresponding to X^n

We can represent sequences X^n using bits $nH(X)$ on average, which is a data compression, since n symbols are represented by $nH(X)$ bits $< n \log |\mathcal{X}|$

Application of AEP to Data Compression

□ Interpretation—how to do this

We divide all sequences in $\mathcal{X}^n$ into two sets the typical set $A_{\epsilon}^{(n)}$ and its complement.

- How many bits we need to encode the sequences in $A_{\epsilon}^{(n)}$?

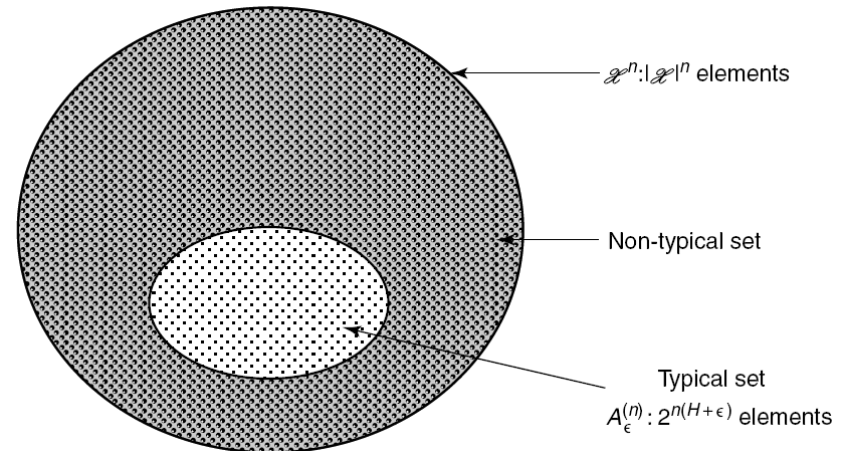
$$\lceil n(H(X) + \epsilon) \rceil < n(H(X) + \epsilon) + 1$$

- How many bits we need to encode the sequences in its complement?

$$\leq n \log |\mathcal{X}| + 1$$

- One additional bit to indicate if we are sending sequence in the typical set or outside.

$$3. |A_{\epsilon}^{(n)}| \leq 2^{n(H(X) + \epsilon)},$$





Application of AEP to Data Compression

$$\begin{aligned} E(l(X^n)) &= \sum_{x^n} p(x^n) l(x^n) \\ &= \sum_{x^n \in A_\epsilon^{(n)}} p(x^n) l(x^n) + \sum_{x^n \in A_\epsilon^{(n)c}} p(x^n) l(x^n) \\ &\leq \sum_{x^n \in A_\epsilon^{(n)}} \underline{\underline{p(x^n)(n(H + \epsilon) + 2)}} + \sum_{x^n \in A_\epsilon^{(n)c}} p(x^n)(n \log |\mathcal{X}| + 2) \\ &= \Pr \{ A_\epsilon^{(n)} \} (n(H + \epsilon) + 2) + \Pr \{ A_\epsilon^{(n)c} \} (n \log |\mathcal{X}| + 2) \\ &\leq n(H + \epsilon) + \epsilon n(\log |\mathcal{X}|) + 2 + 2\epsilon \quad \begin{array}{l} \text{1. } \Pr \{ A_\epsilon^{(n)} \} > 1 - \epsilon \\ \text{2. } \Pr \{ A_\epsilon^{(n)} \} > 1 - \epsilon \end{array} \\ &= n(H + \epsilon'), \quad \epsilon' = \epsilon + \epsilon \log |\mathcal{X}| + \frac{2}{n} \end{aligned}$$



Application of AEP to Data Compression

Theorem 3.2.1 *Let X^n be i.i.d. $\sim p(x)$. Let $\epsilon > 0$. Then there exists a code that maps sequences x^n of length n into binary strings such that the mapping is one-to-one (and therefore invertible) and*

$$E\left[\frac{1}{n}l(X^n)\right] \leq H(X) + \epsilon$$

for n sufficiently large.

□ What is the problem?

- No structure (没有给出如何构造编码的方法) – complexity
- The choice of n (渐进表达式, 理论上无穷) – length (time, storing space)



Application of AEP to Data Compression

□ To evaluate the coding length

■ Example

$$\begin{pmatrix} X \\ p(x) \end{pmatrix} = \begin{pmatrix} x_1 & x_2 & x_3 & x_4 \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{8} & \frac{1}{8} \end{pmatrix}$$

Efficiency 偏离 $H(X)$ 的大小 95%

Not in typical set 10^{-6} , $n=??$

Chebyshev inequality

$$\forall \delta > 0, p[|Y - \mu| \geq \delta] \leq \frac{\sigma^2}{\delta^2} \triangleq \varepsilon, \text{ where } \sigma^2 \triangleq \text{Var}[Y]$$

$$Y = -\frac{\log p(X_1 \cdots X_n)}{n}$$



Application of AEP to Data Compression



$$\forall \delta > 0, p \left[\left| -\frac{\log p(X_1 \cdots X_n)}{n} - H(X) \right| \geq \delta \right] \leq \frac{\sigma^2}{\delta^2} \triangleq \varepsilon,$$

$$\text{w here } \sigma^2 \triangleq \text{Var} \left[-\frac{\log p(X_1 \cdots X_n)}{n} \right]$$

$$\begin{pmatrix} X \\ p(x) \end{pmatrix} = \begin{pmatrix} x_1 & x_2 & x_3 & x_4 \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{8} & \frac{1}{8} \end{pmatrix}$$

$$H(X) = -\sum_{i=1}^4 p_i \log p_i = 1.75$$

$$\frac{H(X)}{H(X) + \delta} = 95\% \Rightarrow \delta \approx 0.092$$

$$\begin{aligned} & \text{Var} \left[\frac{-\log p(X_1 \cdots X_n)}{n} \right] \\ &= \frac{1}{n^2} \text{Var} \left(-\sum_n \log p(X_i) \right) \\ &= \frac{1}{n} \text{Var} (-\log p(X)) = \frac{1}{n} \frac{11}{16} = \frac{1}{n} 0.6875 \end{aligned}$$



Application of AEP to Data Compression

$$\overline{P_e} = \frac{\sigma^2}{\delta^2} = \frac{\text{Var}(-\log p(X))}{n\delta^2} = 10^{-6}$$

$$\Rightarrow n = \frac{\text{Var}}{\delta^2 \overline{P_e}} = \frac{0.6875}{0.092^2 \times 10^{-6}} \cong 0.813 \times 10^8$$

Too large to use!!



Data Compression

- Applying AEP, we know that if $n \rightarrow \infty$, sequences can be compressed up to entropy $nH(X)$.
 - AEP coding
 - Problem: $n \rightarrow \infty$
- Practical coding
- Basic idea
 - Assign short descriptions to the most frequent outcomes of the data source, and necessarily longer descriptions to the less frequent outcomes.
 - 对于出现频率高的输出分配较短的码长，而对于出现频率低的输出分配较长的码长

Some Examples

- Consider a simple example of a code for a random variable:

$$\Pr(X = 1) = \frac{1}{3}, \quad \text{codeword } C(1) = 0$$

$$\Pr(X = 2) = \frac{1}{3}, \quad \text{codeword } C(2) = 10$$

$$\Pr(X = 3) = \frac{1}{3}, \quad \text{codeword } C(3) = 11.$$

- Morse code (摩斯码)**: The Morse code is a reasonably efficient code for the English alphabet using an alphabet of *four symbols: a dot, a dash, a letter space, and a word space*.

Short sequences represent frequent letters (e.g., E)
long sequences represent infrequent letters (e.g., Q)

Alp	code	Alp	Code	Alp	Code	Alp	Code
A	. —	B	— . . .	C	— . —.	D	— . .
E	.	F	. . —.	G	— —.	H	
I	. .	J	. — — —	K	— . —	L	. — . .
M	— —	N	— .	O	— — —	P	. — —.
Q	— — . —	R	. — .	S	. . .	T	—
U	. . —	V	. . . —	W	. — —	X	— . . —
Y	— . — —	Z	— — . .				



Some Examples

□ Example Revisit

$$p(x) = \begin{pmatrix} x_1 & x_2 & x_3 & x_4 \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{8} & \frac{1}{8} \\ 0 & 10 & 110 & 111 \end{pmatrix}$$

$$\bar{n} = \sum_{i=1}^4 p_i n_i = \frac{1}{2} \times 1 + \frac{1}{4} \times 2 + 2 \times \frac{1}{8} \times 3 = 1.75$$

$$H(X) = -\sum_{i=1}^4 p_i \log p_i = 1.75$$

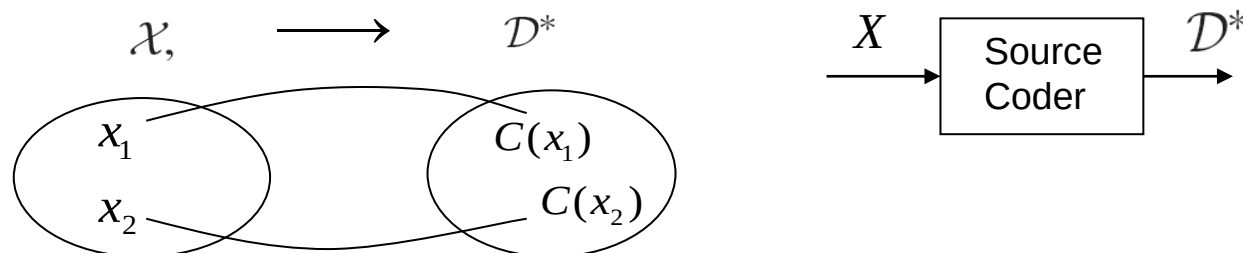
Now let's investigate the variable-length code

Definitions

- A **source code** C for a random variable X is a mapping from $\mathcal{X}$, the range of X , to $\mathcal{D}^*$, the set of **finite-length** strings of symbols from a D -ary alphabet.

$C(x)$ denote the **codeword** corresponding to x

$l(x)$ denote the length of $C(x)$.



The **expected length** $L(C)$ of a source code $C(x)$ for a random variable X with probability mass function $p(x)$ is given by

$$L(C) = \sum_{x \in \mathcal{X}} p(x)l(x),$$



Definitions

□ Non-singular codes

- A code is said to be nonsingular if every element of the range of X maps into a different string in $\mathcal{D}^*$; that is,

$$x \neq x' \Rightarrow C(x) \neq C(x')$$

- 非奇异编码, “不会撞车”
- Is non-singular enough ?



$$C(x_1) = 0$$

$$C(x_2) = 1$$

$$C(x_3) = 01$$

001101-----

酿酒缸缸好做醋坛坛酸

养猪大如山老鼠头头死



Definitions

□ Uniquely decodable codes (唯一可解码)

- Any encoded string comes from a unique sequences of source symbols. (每一个编码序列来自于唯一的信源符号序列)
- For any sequence of source symbols, that sequence can be reconstructed unambiguously from the encoded bit sequence.

X	Singular	Nonsingular, But Not Uniquely Decodable
1	0	0
2	0	010
3	0	01
4	0	10



Definitions

□ Extension (扩展)

- Extension C^* of a code C is the mapping from finite-length strings of $\mathcal{X}$, to finite-length strings of $\mathcal{D}$, defined by

$$C(x_1x_2 \cdots x_n) = C(x_1)C(x_2) \cdots C(x_n).$$

concatenation of codewords. 码字的级联

- A code is uniquely decodable if **its extension is nonsingular**.
- Decoding the output from a uniquely-decodable code, and even determining whether it is uniquely decodable, can be quite complicated. (确定一个编码是不是唯一可解的, 以及进行解码, 都是可能非常困难的)
- A simple class of uniquely decodable codes called **prefix-free** codes. They have many advantages over other uniquely-decodable codes. (幸运的是, 有一类“异前缀码”具有良好的性质)



Definitions

□ prefix-free code 异前缀码

即时码

- A code is called a **prefix-free** code or an **instantaneous** code if no codeword is a prefix of any other codeword.

$C(x_1) = 0$ 没有任何码字是其他码字的前缀

$C(x_2) = 10$ 010111110101100010...

$C(x_3) = 110$

三条很好的性质

$C(x_4) = 111$

- The decoder can decode each codeword of a prefix-free code **immediately** on the arrival of the last bit in that codeword. 译码快
- If a uniquely-decodable code exists with a certain set of codeword lengths, then a prefix-free code can easily be constructed with the same set of lengths. (唯一可解码存在, 则异前缀码存在)
- Given a probability distribution on the source symbols, it is easy to construct a prefix-free code of minimum expected length. 容易构建

Relationship

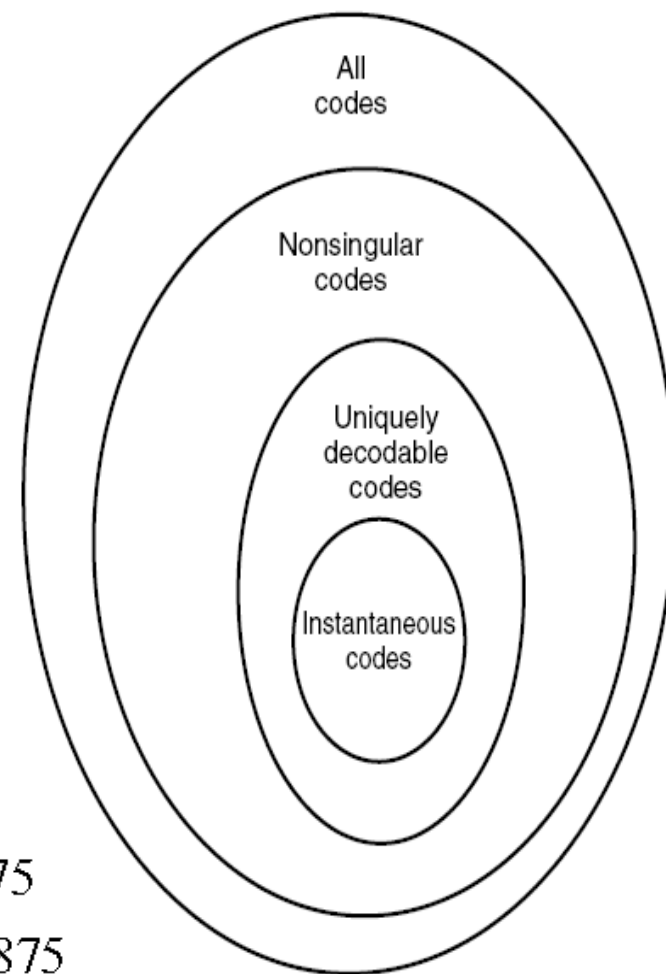
□ relationship

X	Singular	Uniquely Decodable, But Not Instantaneous	
		Instantaneous	
1	0	10	0
2	0	00	10
3	0	11	110
4	0	110	111

$$\begin{aligned}
 C(x_1) &= 0 & C'(x_1) &= 0 \\
 C(x_2) &= 10 & C'(x_2) &= 01 \\
 C(x_3) &= 110 & C'(x_3) &= 011 \\
 C(x_4) &= 111 & C'(x_4) &= 0111
 \end{aligned}$$

$$\bar{n} = 0.5 \times 1 + 0.25 \times 2 + 0.125 \times 3 + 0.125 \times 3 = 1.75$$

$$\bar{n}' = 0.5 \times 1 + 0.25 \times 2 + 0.125 \times 3 + 0.125 \times 4 = 1.875$$





Example

X	Singular	Uniquely Decodable, But Not Instantaneous		Instantaneous
1	0	10		0
2	0	00		10
3	0	11		110
4	0	110		111

An instantaneous code is a
selfpunctuating code

0,1,0,1,1,1,1,1010  0,10,111,110,10

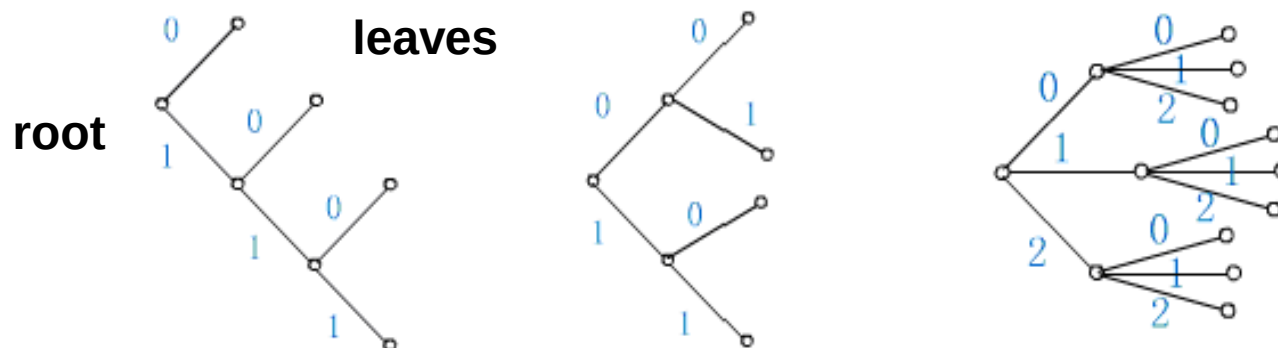
1,1,0,1,1,1,1,00110  434213

An instantaneous code can be decoded without reference to future codewords since the end of a codeword is immediately recognizable

If the first two bits are 00 or 10, they can be decoded immediately. If the first two bits are 11, we must look at the following bits. If the next bit is a 1, the first source symbol is a 3. If the length of the string of 0's immediately following the 11 is odd, the first codeword must be 110 and the first source symbol must be 4; if the length of the string of 0's is even, the first source symbol is a 3. By repeating this argument, we can see that this code is uniquely decodable.

Definitions

□ Code tree (码树)



- A corresponding binary code tree can be defined which grows from a root on the left to leaves on the right representing codewords. (从左边的树根到右边的树叶)
- Each branch is labelled 0 or 1 and each node represents the binary string corresponding to the branch labels **from the root to that node**. (从根到叶，每一个分叉标注0或者1)
- The tree is extended just enough to include each codeword.

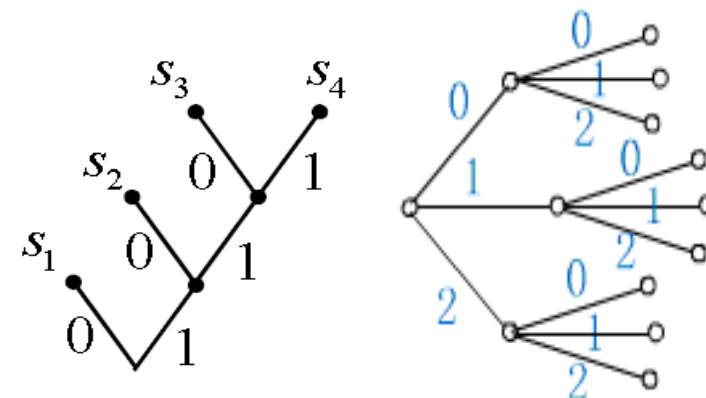
Definitions

□ Relationship

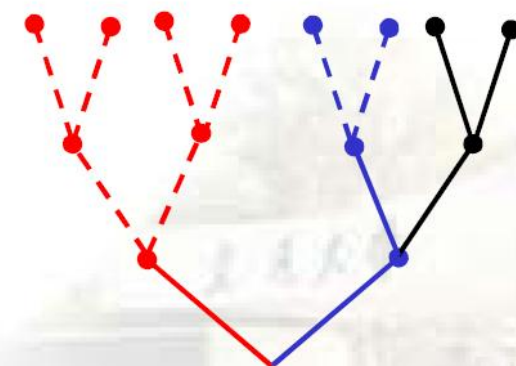
- Root → codeword beginning
- # of branches at each node → #-ary
- Node → part of a codeword
- Leaves → a codeword
- Full tree → fix length code
- Not full tree → variable length code

$$\# \text{ of nodes of a full tree} = \sum_{n=1}^N D^n$$

N Length of a codeword



The prefix free condition implies that no codeword is an ancestor (祖先) of any other codeword on the tree. Hence, each codeword eliminates its descendants (后代) as possible codewords.





Kraft Inequality

- We wish to construct **instantaneous codes** of **minimum expected length** to describe a given source.
- *It is clear that we cannot assign short codewords to all source symbols and still be prefix-free. Then what is the best we can do?*
- **Kraft Inequality**

Theorem 5.2.1 (*Kraft inequality*) *For any instantaneous code (prefix-free code) over an alphabet of size D , the codeword lengths $l_1, l_2, \dots, l_m$ must satisfy the inequality*

$$\sum_i D^{-l_i} \leq 1.$$

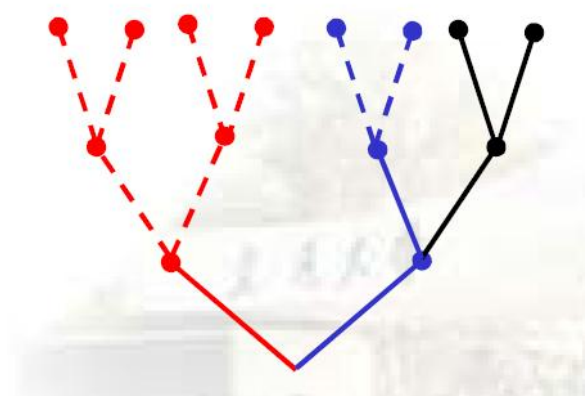
Conversely, given a set of codeword lengths that satisfy this inequality, there exists an instantaneous code with these word lengths.

Kraft Inequality

Proof:

Let $l_{\max}$ be the length of the longest codeword of the set of codewords. Consider all nodes of the tree at level $l_{\max}$. Some of them are codewords, some are descendants of codewords, and some are neither. A codeword at level l_i has $D^{l_{\max}-l_i}$ descendants at level $l_{\max}$. Each of these descendant sets must be disjoint. Also, the total number of nodes in these sets must be less than or equal to $D^{l_{\max}}$. Hence, summing over all the codewords, we have

$$\sum D^{l_{\max}-l_i} \leq D^{l_{\max}} \Rightarrow \sum D^{-l_i} \leq 1,$$



Kraft Inequality

- Comments on Kraft Inequality
 - Any codeword set that satisfies the prefix-free condition has to satisfy the Kraft inequality. (necessary condition)
 - a prefix-free code $\rightarrow$ Kraft inequality holds
 - Kraft inequality is a sufficient condition for the **existence** of a codeword set with the specified set of codeword lengths.
 - Kraft inequality $\rightarrow$ we can find a prefix-free code
 - A code satisfies Kraft inequality $\rightarrow$ it is a prefix-free code

Theorem 5.5.1 (McMillan) *The codeword lengths of any uniquely decodable D -ary code must satisfy the Kraft inequality*

$$\sum D^{-l_i} \leq 1.$$

Conversely, given a set of codeword lengths that satisfy this inequality, it is possible to construct a uniquely decodable code with these codeword lengths.



Optimal Codes

- Prefix-free code with the minimum expected length.
 - Prefix-free code $\rightarrow$ codeword length $l_1, l_2, \dots, l_m$ satisfy Kraft inequality
 - The expected length $L = \sum p_i l_i$ is less than the expected length of any other prefix-free code.

This is a standard optimization problem: Minimize

$$L = \sum p_i l_i$$

over all integers $l_1, l_2, \dots, l_m$ satisfying

$$\sum D^{-l_i} \leq 1.$$

We neglect the integer constraint on $l_1, l_2, \dots, l_m$ and assume equality in the constraint.



Optimal Codes

- Hence, we can write the constrained minimization using **Lagrange multipliers** as the minimization of

$$J = \sum_i p_i l_i + \lambda \left(\sum D^{-l_i} - 1 \right)$$

Differentiating with respect to l_i , we obtain

$$\left\{ \begin{array}{l} \frac{\partial J}{\partial l_i} = p_i - \lambda D^{-l_i} \log_e D = 0 \Rightarrow D^{-l_i} = \frac{p_i}{\lambda \log_e D} \\ \frac{\partial J}{\partial \lambda} = \sum D^{-l_i} - 1 = 0 \Rightarrow \lambda = 1 / \log_e D, \end{array} \right\} p_i = D^{-l_i},$$

$$\boxed{l_i^* = -\log_D p_i.} \quad L^* = \sum p_i l_i^* = -\sum p_i \log_D p_i = H_D(X).$$

But since the codewords length must be integers, we will not always be able to set the codeword lengths as the set.

we should choose a set of codeword lengths “close” to the optimal set.



Optimal Codes

Theorem 5.3.1 *The expected length L of any instantaneous D -ary code for a random variable X is greater than or equal to the entropy $H_D(X)$; that is,*

$$L \geq H_D(X),$$

with equality if and only if $D^{-l_i} = p_i$.

□ Proof

$$\begin{aligned} L - H_D(X) &= \sum_i p_i l_i + \sum_i p_i \log_D p_i \\ &= -\sum_i p_i \log_D D^{-l_i} + \sum_i p_i \log_D p_i \\ &= -\sum_i p_i \log_D \frac{D^{-l_i}}{p_i} \end{aligned}$$

Jensen's inequality

$$\geq -\log_D \sum_i D^{-l_i} \geq 0$$

Kraft inequality



Optimal Codes

Theorem 5.4.1 Let $l_1^*, l_2^*, \dots, l_m^*$ be optimal codeword lengths for a source distribution $\mathbf{p}$ and a D -ary alphabet, and let L^* be the associated expected length of an optimal code ($L^* = \sum p_i l_i^*$). Then

$$H_D(X) \leq L^* < H_D(X) + 1.$$

Proof: Let $l_i = \lceil \log_D \frac{1}{p_i} \rceil$. Then l_i satisfies the Kraft inequality and from

$$\left. \begin{aligned} \sum D^{-\lceil \log \frac{1}{p_i} \rceil} &\leq \sum D^{-\log \frac{1}{p_i}} = \sum p_i = 1. \\ \log_D \frac{1}{p_i} &\leq l_i < \log_D \frac{1}{p_i} + 1. \end{aligned} \right\} \begin{aligned} &\text{we have} \\ &H_D(X) \leq L = \sum p_i l_i < H_D(X) + 1. \end{aligned}$$

There is an overhead that is at most 1 bit (1bit额外开销)

We can reduce the overhead per symbol by spreading it out over many symbols. (通过在多个符号的编码上“摊薄”这一额外的开销)



Optimal Codes

- Consider n consequent symbols to be a supersymbol

$$L_n = \frac{1}{n} \sum p(x_1, x_2, \dots, x_n) l(x_1, x_2, \dots, x_n) = \frac{1}{n} El(X_1, X_2, \dots, X_n).$$

$$H(X_1, X_2, \dots, X_n) \leq El(X_1, X_2, \dots, X_n) < H(X_1, X_2, \dots, X_n) + 1.$$

i. i. d case

$$H(X) \leq L_n < H(X) + \frac{1}{n}.$$

By using large block lengths we can achieve an expected codelength per symbol arbitrarily close to the entropy.

What if for a sequence of symbols from a stochastic process that is not necessarily i.i.d ?

Optimal Codes

- If the stochastic process is stationary, we have the following theorem:

Theorem 5.4.2 *The minimum expected codeword length per symbol satisfies*

$$\frac{H(X_1, X_2, \dots, X_n)}{n} \leq L_n^* < \frac{H(X_1, X_2, \dots, X_n)}{n} + \frac{1}{n}.$$

Moreover, if $X_1, X_2, \dots, X_n$ is a stationary stochastic process,

$$L_n^* \rightarrow H(\mathcal{X}),$$

where $H(\mathcal{X})$ is the entropy rate of the process.



Optimal Codes -- Huffman Code

- So far, we have discussed many properties of the optimal codes.
 - Prefix-free code
 - minimum expected length
- The problem is, how to construct such an optimal code ?

Huffman Code



Huffman Code

□ Huffman procedure

Procedure:

- (1) Arrange the M message in a descent order according to their probabilities. (根据概率高低从高到低排序)
- (2) Assign codeword '0' and '1' to the two messages with the least probabilities, and combine these two messages into a new message with the probability the sum of their probabilities. 分配+合并
- (3) Rearrange the new messages in descent order according to their new probabilities. If the new message has the same probability with another message, put the combined message above. 重排
- (4) Take the procedure iteratively until all the messages are combined 重复2-3



Huffman Code

□ examples $\mathbf{X} : \left\{ \begin{matrix} x_1, & x_2, & x_3, & x_4, & x_5 \\ P(\mathbf{X}) : & 0.4, & 0.1, & 0.2, & 0.2, & 0.1 \end{matrix} \right\}$

message	Probability	Huffman Coding	binary code	l_i
x_1	0.4	1	11	2
x_3	0.2	1	01	2
x_4	0.2	0 (0.4)	00	2
x_2	0.1	1 (0.6)	101	3
x_5	0.1	0 (0.2)	100	3

$$L = \sum_{k=1}^K P(s_k) n_k$$

$$= 0.4 \times 2 + 0.2 \times 2 \times 2 + 0.1 \times 3 \times 2$$

$$= 2.2$$

$$\eta = \frac{H(\mathbf{X})}{L \log D}$$

$$= \frac{2.122}{2.2 \log 2}$$

$$= 96.4\%$$

$$H(\mathbf{X}) = -\sum_{k=1}^K P(s_k) \log P(s_k) = -0.4 \log 0.4 - 2 \times 0.2 \log 0.2 - 2 \times 0.1 \log 0.1 = 2.122$$



Huffman Code

□ Examples ($D>2$ case)

message	Probability	Huffman Coding	binary code	l_i
x_1	0.25	1	1	1
x_3	0.25	2	2	1
x_4	0.2	0	00	2
x_2	0.15	1 0	01	2
x_5	0.15	2 0.5	02	2



Huffman Code

□ Examples ($D>2$ case)

message	Pr	Huffman coding	Code
x_1	0.3	0	00
x_2	0.2	1	01
x_3	0.15	2	02
x_4	0.15	0	10
x_5	0.1	1	11
x_6	0.1	2	12

Note: The Huffman coding column shows a tree structure where the root splits into 0 and 1. The 0 branch splits into 1 and 2. The 1 branch splits into 0 and 1. The 2 branch splits into 0 and 1. The final codes are 00, 01, 02, 10, 11, 12.

Do we get the optimal
code ?

No!

Huffman Code



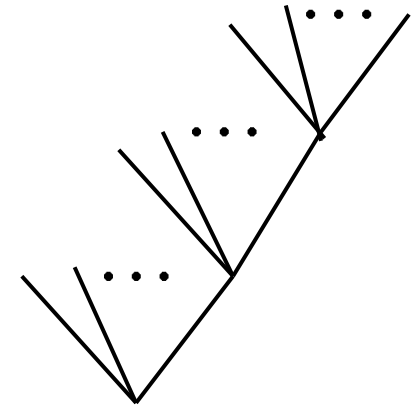
□ $D > 2$ case

- For a D -ary code, the number of prefix-free tree has at most $M = D + m(D-1)$ leaves, where m is the number of the nodes (root is excluded) of the tree.

Proof: $m = 0, \quad M = D$

m Increases by 1, $(D-1)$ new leaves

So we have $M = D + m(D-1)$



In the above example, $N = 6, 1 < m < 2$

All the nodes should be used below the leaves, i.e., only leaves can not be fully used. Otherwise, the code can not be optimal.

So we have to calculate the **empty leaves** before coding.

$$N + B = D + m(D-1) \Rightarrow B = D - N + m(D-1), \quad m = m_{\min} \text{ to make } B \text{ positive.}$$



Huffman Code

□ example

$$B = D - N + m(D - 1),$$

Calculate the empty leaves before coding

$$m_{\min} = 2$$

$$B = 1$$

message	Pr	Huffman coding	Codeword	l_i
x_1	0.3	1	1	1
x_2	0.2	2	22	2
x_3	0.15	1	21	2
x_4	0.15	0	20	2
x_5	0.1	1	01	2
x_6	0.1	0	00	2



Huffman Code

- Huffman code is an optimal (shortest expected length) prefix-free code for a given distribution
 - There are many optimal codes 不唯一 – (inverting all the bits)
 - Huffman procedure constructs one such optimal code by a simple algorithm

Lemma 5.8.1 *For any distribution, there exists an optimal instantaneous code (with minimum expected length) that satisfies the following properties:*

1. *The lengths are ordered inversely with the probabilities (i.e., if $p_j > p_k$, then $l_j \leq l_k$).*
2. *The two longest codewords have the same length.*
3. *Two of the longest codewords differ only in the last bit and correspond to the two least likely symbols.*



Huffman Code

- If $p_j > p_k$, then $l_j \leq l_k$. Here we swap codewords.
Consider C'_m , with the codewords j and k of C_m interchanged. Then

$$\begin{aligned} L(C'_m) - L(C_m) &= \sum p_i l'_i - \sum p_i l_i \\ &= p_j l_k + p_k l_j - p_j l_j - p_k l_k \\ &= (p_j - p_k)(l_k - l_j). \end{aligned}$$

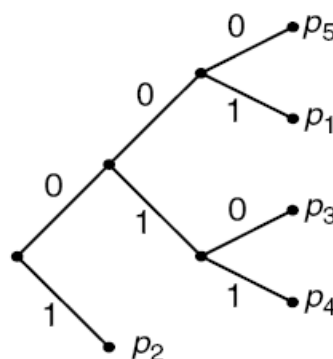
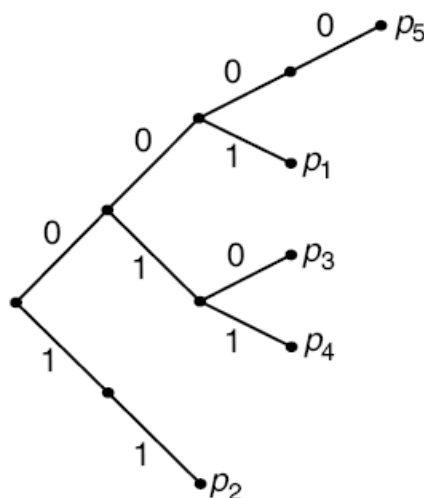
But $p_j - p_k > 0$, and since C_m is optimal, $L(C'_m) - L(C_m) \geq 0$.
Hence, we must have $l_k \geq l_j$. Thus, C_m itself satisfies property 1.

In fact, this property is true for all optimal codes (stronger than the “there exists” statement of the theorem).

Huffman Code

修剪; 切除

- The two longest codewords are of the same length.* Here we trim the codewords. If the two longest codewords are not of the same length, one can delete the last bit of the longer one, preserving the prefix property and achieving lower expected codeword length. Hence, the two longest codewords must have the same length. By property 1, the longest codewords must belong to the least probable source symbols.

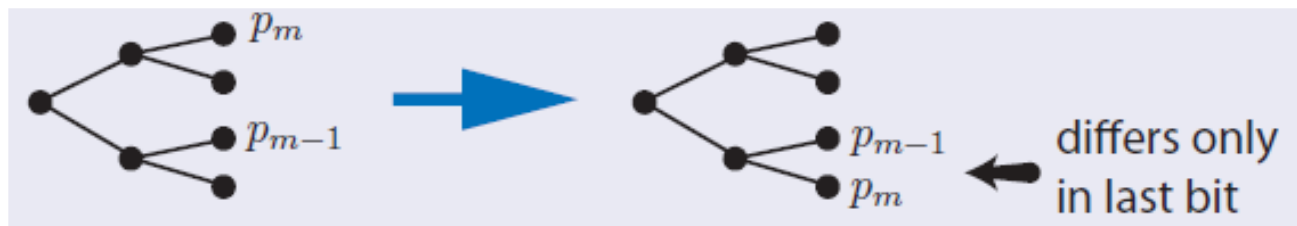


Optimal code must have two longest codewords with the same length.

Huffman Code

- *The two longest codewords differ only in the last bit and correspond to the two least likely symbols.* Not all optimal codes satisfy this property, but by rearranging, we can find an optimal code that does. If there is a maximal-length codeword without a sibling, we can delete the last bit of the codeword and still satisfy the prefix property. This reduces the average codeword length and contradicts the optimality of the code. Hence, every maximal-length codeword in any optimal code has a sibling. Now we can exchange the longest codewords so that the two lowest-probability source symbols are associated with two siblings on the tree. This does not change the expected length, $\sum p_i l_i$. Thus, the codewords for the two lowest-probability source symbols have maximal length and agree in all but the last bit.

兄弟姊妹





Huffman Code

Summarizing, we have shown that if $p_1 \geq p_2 \geq \dots \geq p_m$, there exists an optimal code with $l_1 \leq l_2 \leq \dots \leq l_{m-1} = l_m$, and codewords $C(x_{m-1})$ and $C(x_m)$ that differ only in the last bit. $\square$

Theorem 5.8.1 *Huffman coding is optimal; that is, if C^* is a Huffman code and C' is any other uniquely decodable code, $L(C^*) \leq L(C')$.*

We omit the proof here.



Summary

□ The goal of source coding – efficiency

□ AEP for Data Compression

$$E\left[\frac{1}{n}l(X^n)\right] \leq H(X) + \varepsilon$$

■ AEP for coding

■ Non structure

■ Length

□ Code tree

□ Prefix-free code

□ Kraft Inequality $\sum D^{-l_i} \leq 1.$

■ A prefix-free code $\rightarrow$ Kraft inequality holds

■ A code satisfies Kraft inequality $\rightarrow$ it is a prefix-free code

Summary

□ Optimal Codes

- Prefix-free code with the minimum expected length

- For iid

$$H_D(X) \leq L^* < H_D(X) + 1.$$

- For random process

$$\frac{H(X_1, X_2, \dots, X_n)}{n} \leq L_n < \frac{H(X_1, X_2, \dots, X_n)}{n} + \frac{1}{n}.$$

- For stationary process

$$L_n \rightarrow H(\mathcal{X}).$$

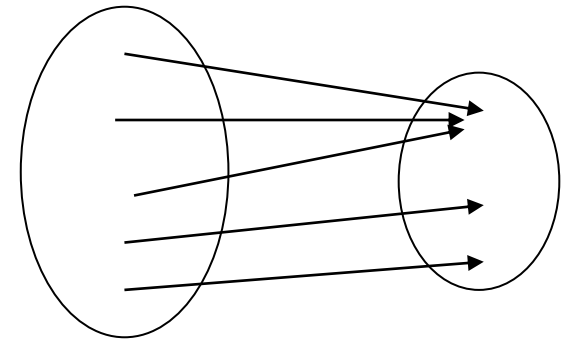
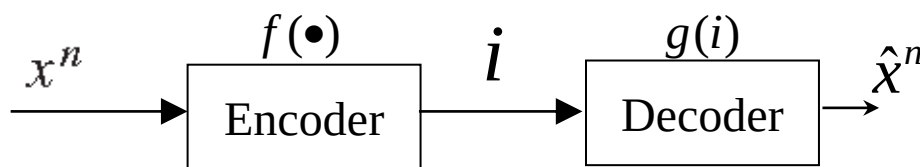
□ Huffman Code——optimal code

- $D = 2$;
- $D > 2$;

Application of AEP to Data Compression

□ Random coding (随机编码)

- Assign each sequence $x^n = x_1, x_2, \dots, x_n$ to an index i drawn uniformly from $i \in \{1, 2, \dots, 2^{nR}\}$



- Problem: less indices than sequences $\rightarrow$ error
- Decoding procedure: AEP decoding (AEP解码)
 $g(i) \equiv$ the unique x^n such that $f(x^n) = i$ and $x^n \in A_\epsilon^{(n)}$

Now, what is the error rate?

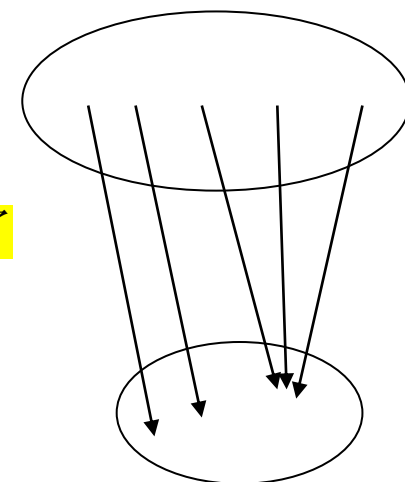
Application of AEP to Data Compression

□ Error rate (错误概率分析)

$$P(e) = P\left[\underbrace{\{x^n \notin A_\varepsilon^{(n)}\}}_{\text{Not in typical set but assigned a index 给非典型序列分配了编号}} \cup \underbrace{\{\tilde{x}^n \in A_\varepsilon^{(n)} : f(x^n) = f(\tilde{x}^n)\}}_{P(E) \text{ In typical set, but not unique 是典型序列, 但是“撞车”了}}\right]$$

Not in typical set
but assigned a index
给非典型序列分配了
编号

In typical set, but not unique
是典型序列, 但是“撞车”了



$$\begin{aligned} P(E) &= \sum_{\substack{\tilde{x}^n \in A_\varepsilon^{(n)} \\ \tilde{x}^n \neq x^n}} P(f(x^n) = f(\tilde{x}^n)) = \sum_{\substack{\tilde{x}^n \in A_\varepsilon^{(n)} \\ \tilde{x}^n \neq x^n}} 2^{-nR} \\ &\leq |A_\varepsilon^{(n)}| 2^{-nR} \leq 2^{n(H(X) + \varepsilon)} 2^{-nR} = 2^{-n(R - H(X) - \varepsilon)} \end{aligned}$$

$$P(e) \leq \varepsilon + 2^{-n(R - H(X) - \varepsilon)} \xrightarrow{n \rightarrow \infty} 0, \quad \text{if } R > H(X)$$



Application of AEP to Data Compression

□ Comparison

■ AEP encoder

- Fix-to-variable length coder
(Different sequences have
codewords of different length)

$$P(e) = 0$$

$$\frac{E(l(x^n))}{n} \rightarrow H(X)$$

- AEP used at encoder
(we need distribution, or typical
set at encoder)

■ Random enc, AEP dec

- Fix-to-Fix length coder
(All codewords length nR where
 $R > H(X)$)

$$P(e) \rightarrow 0$$

$$\frac{1}{n} l(x^n) = R$$

- AEP used at decoder
(we need distribution, or typical
set at the decoder)